Physics 411 Assignment 1

Problem 1

The following are the two's complements for the following decimal numbers in 16 bits:

- 10: 0000 0000 0000 1010

In Binary, 10 is 1010, and left most value is 0 as it is positive.

- 436: 0000 0001 1011 0100

In Binary, 10 is 110110100, and left most value is 0 as it is positive.

- 1024: 0000 0100 0000 0000

In Binary, 10 is 100 0000 0000, and left most value is 0 as it is positive.

- -13: 1111 1111 1111 0011

In Binary 13 is 1101. In 16 bits, it is 0000 0000 0000 1101. One's complement is 1111 1111 1111 0010, and when we add 1, we get 1111 1111 1111 0011 for -13.

- -1023: 1111 1100 0000 0001

In Binary, 1023 is 11111111. In 16 bits, it is 0000 0011 1111 1111. One's complement is 1111 1100 0000 0000, and when we add 1, we get 1111 1100 0000 0001 for -1023.

- -1024: 1111 1100 0000 0000

In Binary, 1024 is 10000000000. In 16 bits, it is 0000 0100 0000 0000. Its one's complement is 1111 1011 1111 1111. When we add 1, we get 1111 1100 0000 0000 for -1024

Problem 2

For an int:

An int stores 16 bits. As it is a signed data type, it can store up to $2^{15} - 1$ value, which evaluates to 32,767.

If we follow the function, $f_0 = 2, f_1 = 4, f_2 = 16, f_3 = 256, f_4 = 65{,}536$. As this value is more than the largest possible value for an int data type, maximum value of n is 3.

For a long int:

A long int stores 32 bits. As it is a signed data type, it can store up to $2^{31} - 1$ value, which evaluates to 2,147,483,647.

If we follow the function, $f_0 = 2, f_1 = 4, f_2 = 16, f_3 = 256, f_4 = 65{,}536, f_5 = 4{,}294{,}967{,}296$ As this value is more than the largest possible value for a long int data type, maximum value of n is 4.

For an unsigned long int:

A long int stores 32 bits. As it is a signed data type, it can store up to $2^{32} - 1$ value, which evaluates to 4,294,967,295.

If we follow the function, $f_0 = 2, f_1 = 4, f_2 = 16, f_3 = 256, f_4 = 65{,}536, f_5 = 4{,}294{,}967{,}296$ As this value is more than the largest possible value for an unsigned long int data type, maximum value of n is still 4.